

VIABILITY OF COST OF CARRY MODEL FOR ORCHARD FORWARDS IN KASHMIR

INTRODUCTION

Kashmiri apples and dry fruits are famous all over the world. In Kashmir large number of farmers earn their livelihood from apples and other orchards. They always pray for good harvest and high market price.

Farmers in Kashmir lack professional knowledge of finance. Without knowing the fact most of the orchard owners sell the complete orchards in the forward market for one, two or maximum for three years and the buyer (mostly dealers) of the orchard take the responsibility to maintain the orchard. The paper analyses the applicability of the traditional cost of carry model of forward pricing for orchard forward contract. Further, an attempt is made to answer one important research question that is how to calculate the forward price in case of orchard forwards (forward in which complete orchard, whether it is apple, almond or walnut orchard, is an underlying asset) particularly in Kashmir and in other orchards in general.

Further, the paper makes an attempt to present a modified model to make it applicable in case of the orchards in Kashmir in particular and orchards in other parts of the world in general.

The paper is divided into the following sections:

1. Literature Review and Research Methodology.
2. Various ways of selling the harvest of orchards in Kashmir.
3. Problems in application of cost of carry model in case of orchards.
4. Modifications in the cost of carry model to make it applicable in case of pricing orchards.
5. Limitation of the proposed model and future research.

1. LITERATURE REVIEW

Chicago Board of Trade first started the future contract in 1848, Chow et.al. (2000) otherwise the history of forward contracts is obscure. In forward contracts one party hedges to reduce risk and other party enter into the contract to earn profit. Forward contracts help to minimize risks of small farmers. Forward Contract is an agreement between two parties in which one party agrees to buy or sell the underlying on a specified future date and at a specified price. Two popular theories of forward pricing are the Risk Premium model and Cost of Carry Model, Fama & French (1987). Cost of Carry model, also known as theory of storage, has been adopted by researchers like Brennan (1958), Deaton and Laroque (1992), Kaldor (1939), Telser (1958),

and Working (1948, 1949), wherein Forward price is equal to the spot price plus carrying cost based on arbitrage argument. Forward price is equal to:

$F(0,T) = S_0e^{rt} + \text{Future Value of Carrying Cost} - \text{Future Value of Convenience Yield}$,(Chow el.al.2000). Risk Premium model is popularized by Breeden (1980), Cootner (1939), Dusak (1973), Hicks (1939), Houthakker (1968) and Jaganathan (1985), the model requires data with high frequency which is not available in case of orchard forwards.

So keeping in view the nature of orchard forwards, the paper in its first phase adopted Cost of Carry model.

2. VARIOUS WAYS OF SELLING THE HARVEST IN ORCHARDS IN KASHMIR

Depending upon the size of the orchard and time the orchard owner has, the following are the main schemes of selling the harvest in orchards. These schemes are worked out after interviewing the orchard owners/farmers and dealers in Kashmir.

- I. Farmer/Owner Maintains the orchard: Mostly the small farmers in Kashmir maintain the orchard themselves and incur all the costs associated with the orchard. At the time of harvest they have the following options:
 - i. Sell the harvest in the wholesale market at the local Mandi or outside the state.
 - ii. Sell the harvest or complete orchard at the time of harvest to the broker who sell the harvest in the wholesale market at the local or national mandis.
 - iii. Farmers store the harvest and sell later on.
 - iv. Sell the low quality harvest in the local retail market and good quality product at the local or national mandis.
- II. Farmer/Owner sell the orchard before the harvest in forward market: Mostly the farmers having large orchards sell the orchard before the harvest. The contract may be for one, two or three years. The buyer takes the responsibility to maintain the orchard and incur the complete cost. The cost includes the labour cost, cost of fertilizers and pesticides, cost of marketing, transportation cost etc. The risk of fluctuations in production and market price also shifts from owner to the buyer/dealer. Sometimes dealer pays some of the agreed price in advance to the owner/farmer.

The paper is concerned about the second arrangement where the farmer/owner of the orchard sells the orchard in the forward market. The most important question is how to calculate the forward price of the orchard. Is it possible to apply the traditional cost of carry model in the forward pricing, where the underlying asset is orchard? The paper tries to answer this question.

PROBLEMS IN APPLICATION OF COST OF CARRY MODEL IN CASE OF ORCHARDS

Based on arbitrage, forward price is equal to (Chow et.al., 2000):

$$F(0,T) = S_0e^{rt} + \text{Future Value of Carrying Cost} - \text{Future Value of Convenience Yield.}$$

S_0 is the spot price of the underlying, r is the risk free rate and t is the time to maturity of the forward contract. Carrying cost is the cost of storage and other costs. Convenience yield is the yield of having the inventory so that unexpected demand can be met. In this paper researcher assume that there is no convenience yield as there is no convenience of holding the inventory of orchards which are not yet ready for harvest.

So we require spot rates of the underlying, cost of carry like storage, insurance, transport etc. to calculate forward price or value of forward on any underlying. This price and value is based on no arbitrage principle i.e. if price/value calculated by this model is equal to market price/value than no arbitrage is possible.

Above model is based on the over simplified assumptions. Model assumes that risk free rate of return i.e. the rate at which the parties lend and borrow is same for both the parties, short selling is possible, spot price is available from the market, forward price is the unbiased estimator of the expected spot price, market is efficient, no bid-ask spread, no brokerage, no transaction cost, no cash inflows and outflows from the market and easy to buy and sell the underlying asset.

The above model is used to calculate the forward pricing of financial instruments in the efficient markets. In the inefficient markets like the market of the orchards in Kashmir and other agricultural products in rural areas the assumptions are not valid. Also there is asymmetric information i.e. one party has more knowledge of the market than the other. Most of farmers in rural India are illiterate. Even contractors are ignorant about the forward contracts but with experience they have lot of knowledge about the local market of the agriculture products. Due to their knowledge they earn lot of profit at the cost of the farmer's earnings.

Assumptions of cost of carry model:

1. S_0 is known with certainty
2. r is risk free rate and fixed
3. T is time and fixed
4. Cost to be incurred on the underlying is known and fixed
5. Income/Yield on the underlying is known and fixed
6. No exchange of money at the initiation of the contract
7. Both parties have same information and complete transparency i.e. market is efficient.

Why Cost of Carry Model cannot apply in orchard forwards:

In rural areas the parties both long and short of the agriculture farm/orchard contracts set forward price arbitrarily based on experience. The long party mostly traders have more information about the market than the short party (farmers). This violates one of the very important assumption of cost of carry model of forward pricing. Long has more information about the demand and supply, market price, transportation cost, storage cost etc. of the underlying. Beside this while applying the cost of carry model to calculate the forward price in agriculture farm/orchard as an underlying, parties face following problems:

- I. Spot price is required to calculate the forward price. But the spot price of the underlying is not be available e.g. in case of apples and other fruits and dry fruits the price at the time of contract is not available from the market. Even if the price is available it is very high due to low supply at the time of the initiation of the contract which cannot be used in the model as a spot price. Instead of spot price the estimated future price of the underlying at expiry can be used in the model. For estimating the future price the estimate of the production per hectare and the price of the underlying is required.

Past data can be used to model the price of the underlying at expiry. To calculate the future price at expiry the two estimates are required, one is estimate of production and another is estimate of the price.

- i. **Modelling Production:** Production can be modelled with the help of agriculture scientists. It depends upon many factors like past production, estimated climatic conditions, type of soil, variety, amount of fertilizers, pesticides, rainfall, snowfall, chances of storm etc. So, it is a specialized area for which we require the help of agriculture scientists.

Production per hectare= $\hat{a}$ (past production, rainfall, snowfall, typo of soil, fertilizers, pesticides etc.)

Statistical estimates based on normal distribution of past data can also be used to estimate the production. The interval estimate can be estimated as follows:

$$\text{Production} = X \pm Z \hat{\sigma}$$

X is the average production of the underlying per hectare in the past years, Z score depends upon the confidence interval and level of significance and $\hat{\sigma}$ is the standard deviation of the production per hectare.

The range within which the production of the underlying lies can be calculated with the above model.

- ii. Estimates of price: This can be estimated with the help of inflation forecast at the expiry. Simply multiplying the previous year prices with inflation forecast. The market price of the underlying in the previous year which can be easily obtained from the wholesale mandies is required for estimating the price.

In this manner the range of estimated price at expiry of the orchards can be calculated.

- II. Second problem to apply the model is the estimate of cost of carry up to expiry of the forward contracts. Which includes the cost of labour, fertilizers, pesticides, packaging, transportation etc. The cost of carry per hectare can be estimated by estimating the quantity of inputs required and the price of the inputs. Quantity per hectare of inputs can be estimated with the help of agriculture scientists and experts in the field and price of inputs is available in the market.

Another issue is who pays this price. If price is paid by the farmer i.e. short then it should be added in the cost of carry model and if it is paid by the long it should not be added in the model to make it arbitrage free.

- III. Adjustment of advance payment: In the conventional cost of carry model it is assumed that no money changes hands at the initiation of the contract. But in case of agriculture forwards this is not the case. Long pays the short some amount of the agreed price in advance and rest at the expiry of the contract. Simple adjustment is to subtract the advance amount from the present value of the estimated price of orchard at expiry and compound the balance, which is to be paid at expiry.

Balance forward price: $F(0,T) = (S_0e^{rt} - \text{advance})e^{rt}$

- IV. Adjustment of risk free rate: In the conventional cost of carry model risk free interest rate is used and is assumed to be fixed and both long and short borrow and lend at the same interest rate to make the model arbitrage free. In case of large institutional investors like international banks it holds but not for the farmers and small traders in the orchard market. They borrow at very high rate and lend at very low rate (mostly lend by depositing in the saving bank account). If short receives some advance from long he invests at very low rate and long has to borrow at the very high rate may be at the rate of consumption loan which is very high particularly in India. This problem is difficult to adjust because of differences in borrowing and lending rates.

In order to adjust this in the cost of carry model, we can take the average of borrowing and lending rate for the small investors e.g. banks saving rate for lending and personal

consumption loan rate for borrowing charged by banks. The risk free rate cannot be used in the model.

PROPOSED MODEL

Before calculating the forward price in case of orchard forwards two decisions are required. First is whether at the initiation of the forward contract the advance is paid by the long to the short or not and second is whether the cost is incurred by the short or the long. Based on these decisions following models are proposed to calculate the price of the orchard forward contracts: ($F(0,T)$ is the Forward Price, S_0e^{rt} is the value of the orchard at expiry of the forward contract. S_0e^{rt} and Future value of Cost are estimated as explained above by modelling production per hectare and price).

- i. Without advance and carrying cost paid by the short:
$$F(0,T) = S_0e^{rt} + \text{Future Value of Cost.}$$
- ii. Without advance and carrying cost paid by the long: $F(0,T) = S_0e^{rt}$
- iii. With partial advance by long to the short and carrying cost paid by the short: $F(0,T) = (S_0 - \text{advance})e^{rt} + \text{Future Value of Cost.}$
- iv. With advance by long to the short and carrying cost paid by the long party: $F(0,T) = (S_0 - \text{advance})e^{rt}.$

CONCLUSION

Although the farmers have less knowledge of the market than the contractors/dealers but both farmers and dealers fix the forward price arbitrarily with experience. Here risk shifted from short to the long. This risk can be adjusted in estimating production. One party's gain in the contract is other party's loss. With the help of the proposed model both the parties can estimate the forward price and reduce the magnitude of this loss.

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